

Problem 1

- If n is odd

- Consider x^i for $0 < i < n$ s.t. x is a rotation

$$x^j x^i x^{-j} \text{ for } 0 \leq j < n$$

$$= x^i$$

$$yx^j x^i yx^j \text{ for } 0 \leq j < n$$

$$= yyx^{n-j-i}x^j = x^n x^{-i} x^{j-j} = x^{-i}$$

$$\text{Thus } C_{x^i} = \{x^i, x^{-i}\} = C_{x^{-i}}$$

- Consider y s.t. y is a reflection

$$x^j y x^{-j} \text{ for } 0 \leq j < n$$

$$= y x^{n-j} x^{-j} = y x^{-2j}$$

Since n is odd we can solve $k \equiv -2j \pmod{n}$

$$\text{thus } C_y = \{yx^k \mid k \in \mathbb{Z}\} = \{yx^k \mid k \in \{0, 1, 2, \dots, n-1\}\}$$

$$\text{Thus } |D_n| = \sum_{[x^i]} |C_{x^i}| + |C_y| = |\{e\}| + \sum_{[x^i] \setminus [e]} |C_{x^i}| + |C_y| = 1 + (n-1) + n = 2n$$

- If n is even

- It is the same for rotations

- Consider y s.t. y is a reflection

$$x^j y x^{-j} \text{ for } 0 \leq j < n$$

$$= y x^{n-j} x^{-j} = y x^{-2j}$$

$$\text{thus } C_y = \{y, yx^2, yx^4, \dots, yx^{n/2-1}\}$$

- Consider yx

$$x^j y x x^{-j} \text{ for } 0 \leq j < n$$

$$= y x^{n-j} x^{1-j} = y x^{1-2j}$$

$$\text{thus } C_{yx} = \{yx, yx^3, yx^5, \dots, yx^{n-1}\}$$

$$\text{Thus } |D_n| = \sum_{[x^i]} |C_{x^i}| + |C_y| + |C_{yx}|$$

$$= |\{e\}| + |\{x^{n/2}\}| + \sum_{[x^i] \setminus [e] \cup [x^{n/2}]} |C_{x^i}| + |C_y| + |C_{yx}| = 1 + 1 + 2(n/2 - 1) + 2(n/2) = 2n$$

Problem 2

- Conjugacy class of 1:

$$\{1\}$$

Since 1 is the identity element, it commutes with all other elements:

$$h \cdot 1 \cdot h^{-1} = 1 \quad \text{for all } h \in Q_8.$$

- Conjugacy class of -1 :

$$\{-1\}$$

Similar to 1, the element -1 commutes with all other elements:

$$h \cdot (-1) \cdot h^{-1} = -1 \quad \text{for all } h \in Q_8.$$

- Conjugacy class of i and $-i$:

The conjugates of i by other elements of Q_8 are:

$$1 \cdot i \cdot 1^{-1} = i, \quad (-1) \cdot i \cdot (-1)^{-1} = i,$$

$$j \cdot i \cdot j^{-1} = -i, \quad k \cdot i \cdot k^{-1} = -i.$$

Therefore, the conjugacy class of i is:

$$\{i, -i\}.$$

Similarly, for $-i$:

$$j \cdot (-i) \cdot j^{-1} = i, \quad k \cdot (-i) \cdot k^{-1} = i,$$

so i and $-i$ share the same conjugacy class.

- Conjugacy class of j and $-j$:

The conjugates of j are computed as follows:

$$1 \cdot j \cdot 1^{-1} = j, \quad (-1) \cdot j \cdot (-1)^{-1} = j,$$

$$i \cdot j \cdot i^{-1} = -j, \quad k \cdot j \cdot k^{-1} = -j.$$

Hence, the conjugacy class of j is:

$$\{j, -j\}.$$

Similarly, for $-j$:

$$i \cdot (-j) \cdot i^{-1} = j, \quad k \cdot (-j) \cdot k^{-1} = j,$$

so j and $-j$ are in the same conjugacy class.

- Conjugacy class of k and $-k$:

The conjugates of k are:

$$1 \cdot k \cdot 1^{-1} = k, \quad (-1) \cdot k \cdot (-1)^{-1} = k,$$

$$i \cdot k \cdot i^{-1} = -k, \quad j \cdot k \cdot j^{-1} = -k.$$

Therefore, the conjugacy class of k is:

$$\{k, -k\}.$$

Similarly, for $-k$:

$$i \cdot (-k) \cdot i^{-1} = k, \quad j \cdot (-k) \cdot j^{-1} = k,$$

so k and $-k$ share the same conjugacy class.

Thus, the conjugacy classes of Q_8 are:

$$\{1\}, \quad \{-1\}, \quad \{i, -i\}, \quad \{j, -j\}, \quad \{k, -k\}.$$

The sizes of these conjugacy classes are:

$$|\{1\}| = 1, \quad |\{-1\}| = 1, \quad |\{i, -i\}| = 2, \quad |\{j, -j\}| = 2, \quad |\{k, -k\}| = 2.$$

Therefore, the class equation of Q_8 is:

$$8 = 1 + 1 + 2 + 2 + 2.$$

Problem 3

- (a) • (i. $\implies$ iii.) Assume i.

Let $\varphi : G \rightarrow K$ s.t. $\varphi(g) = \varphi(hk) = k$

Well Defined: Let $g \in G$ by i. since $g = hk$ is unique this implies every input has one output.

Kernel: Let $g \in \ker(\varphi)$

$$\varphi(g) = \varphi(hk) = e \implies k = e \text{ thus } g = h \in H$$

Let $h \in H$ thus $he \in HK$ thus $\varphi(h) = \varphi(he) = e$ thus $H = \ker(\varphi)$

Surjective: Let $k \in K$ consider: $ek \in G$

$\varphi(ek) = k$ thus φ is surjective.

By the first isomorphism theorem this implies $K \cong G/H$

- (iii. $\implies$ ii.) Assume iii. with the map φ

Since φ is injective $\ker(\varphi) = \{e\} \implies \ker(\varphi) \subseteq K \cap H$ Let $l \in K \cap H \implies l \in K$ and $l \in H$

thus $\varphi(l) = lH$ but since $l \in H \implies \varphi(l) = H$ thus $l \in \ker(\varphi)$ thus $\ker(\varphi) = K \cap H = \{e\}$

Since φ is surjective this means for all $kH \in G/H$ there exists a $k' \in K$ s.t. $\varphi(k') = kH$

and since G/H partitions G this implies for all $g \in G \implies g \in jH$ for some $j \in K \implies$

$g = jh$ for some $h \in H$

thus $G = HK$

- (ii. $\implies$ i.) Assume ii.

Let $g \in G$ s.t. $g = hk = ij$ for $h, i \in H$ and $k, j \in K$

thus $i^{-1}h = jk^{-1} \implies$ that is in $H \cap K$ thus $i^{-1}h = e \implies i = h$ and $jk^{-1} = e \implies j = k$

thus all $g \in G$ are a unique product of an $h \in H$ and $k \in K$

- (b) Let $\varphi : G \rightarrow (H \times K, \text{ with the given group multiplication})$ s.t. $\varphi(hk) = (h, k)$

Homomorphism: Let $g_1, g_2 \in G$ thus $g_1 = h_1k_1$ and $g_2 = h_2k_2$ for $h_1, h_2 \in H$ and $k_1, k_2 \in K$

$\varphi(g_1g_2) = \varphi(h_1k_1h_2k_2) = \varphi(h_1(k_1h_2k_1^{-1})k_1k_2)$ since H is normal in G this means $k_1h_2k_1^{-1} \in H$

thus $\varphi(h_1(k_1h_2k_1^{-1})k_1k_2) = (h_1(k_1h_2k_1^{-1}), k_1k_2) = (h_1, k_1)(h_2, k_2) = \varphi(g_1)\varphi(g_2)$

Injective: Let $g \in \ker(\varphi)$ thus $g = hk$ for $h \in H$ and $k \in K$

$\varphi(g) = \varphi(hk) = (h, k) = (e, e)$ thus $h = e$ and $k = e$ thus $g = ee = e$ thus $\ker(\varphi) = \{e\}$

Surjective: Let $(h, k) \in (H \times K, \text{ with the given group multiplication})$ consider $hk = g \in G$

$\varphi(g) = \varphi(hk) = (h, k)$

Thus $G \cong (H \times K, \text{ with the given group multiplication})$

- (c) Let $k \in K$ consider ϕ_k

Homomorphism: Let $h, j \in H$

$$\phi_k(hj) = khjk^{-1} = khk^{-1}kj k^{-1} = \phi_k(h)\phi_k(j)$$

Injective: Let $h \in \ker(\phi_k)$

$$\phi_k(h) = khk^{-1} = e \implies h = e \text{ thus } \ker(\phi_k) = \{e\}$$

Surjective: Let $h \in H$ since H is normal consider $k^{-1}hk$

$$\phi_k(k^{-1}hk) = kk^{-1}hkk^{-1} = h$$

Thus ϕ_k is an automorphism

Homomorphism: Let $k, j \in K$

$$\phi(kj)(h) = \phi_{kj}(h) = kjhj^{-1}k^{-1} = \phi_k \circ \phi_j(h) = \phi(k)\phi(j)(h)$$

Thus ϕ is a homomorphism

If H, K are an internal direct product then $\phi(k) = id_H$ for all $k \in H$

Thus the group law of $H \times K$ would be $(h_1, k_1)(h_2, k_2) = (h_1h_2, k_1k_2)$ thus proven in last homework this means G is a direct product of H, K

Problem 4

- (a)
- (Closure) Given by group law this is closed
 - (Identity) Consider $(e, e), (h, k) \in H \rtimes_{\phi} K$
 $(e, e)(h, k) = (e\phi_e(h), ek)$ since ϕ is a homomorphism and $\phi(e) = \phi_e$ this means that $\phi(e) = \phi_e = id_H$ thus $= (h, k)$
 $(h, k)(e, e) = (h\phi_k(e), ke)$ since ϕ_k is an automorphism it is a homomorphism thus $\phi_k(e) = e$ thus $= (he, k) = (h, k)$
 - (Associativity) Let $(h_1, k_1), (h_2, k_2), (h_3, k_3) \in H \rtimes_{\phi} K$
 $(h_1, k_1)[(h_2, k_2)(h_3, k_3)] = (h_1, k_1)(h_2\phi_{k_2}(h_3), k_2k_3) = (h_1\phi_{k_1}(h_2\phi_{k_2}(h_3)), k_1k_2k_3)$
 $[(h_1, k_1)(h_2, k_2)](h_3, k_3) = (h_1\phi_{k_1}(h_2), k_1k_2)(h_3, k_3) = (h_1\phi_{k_1}(h_2)\phi_{k_1k_2}(h_3), k_1k_2k_3)$
 $= (h_1\phi_{k_1}(h_2\phi_{k_2}(h_3)), k_1k_2k_3)$
 - (Inverses) Let $(h, k) \in H \rtimes_{\phi} K$
Consider $(\phi_{k^{-1}}(h^{-1}), k^{-1})$
 $(h, k)(\phi_{k^{-1}}(h^{-1}), k^{-1}) = (h\phi_k(\phi_{k^{-1}}(h^{-1})), kk^{-1}) = (h\phi_e(h^{-1}), e) = (e, e)$
 $(\phi_{k^{-1}}(h^{-1}), k^{-1})(h, k) = (\phi_{k^{-1}}(h^{-1})\phi_{k^{-1}}(h), k^{-1}k) = (\phi_{k^{-1}}(e), e) = (e, e)$
- (b) The internal and external semidirect products provide two perspectives on how groups can combine through an action. In the external semidirect product, we start with two independent groups H and K and define a homomorphism $\phi : K \rightarrow \text{Aut}(H)$ to describe how K acts on H . This action defines a group structure on the set $H \times K$, where the multiplication is given by $(h_1, k_1) \cdot (h_2, k_2) = (h_1 \cdot \phi(k_1)(h_2), k_1 \cdot k_2)$. The internal semidirect product, on the other hand, starts with a single group G that can be decomposed into two subgroups $H \triangleleft G$ and $K \leq G$. Here, H is a normal subgroup, and K acts on H via conjugation: $khk^{-1} \in H$ for $k \in K$ and $h \in H$. Both constructions describe the same type of group structure, where the elements of one group act on another.
- (c)
- (Identity) Let $\phi \in \text{Hom}(K, \text{Aut}(H))$, and let $(\psi, \mu) = (id_H, id_K)$
 $((id_H, id_K)\phi)(k) = id_H \circ \phi(id_K^{-1}(k)) \circ id_H^{-1}$
 $= id_H \circ \phi(k) \circ id_H = \phi(k)$
 - Let $(\psi_1, \mu_1), (\psi_2, \mu_2) \in \text{Aut}(H) \times \text{Aut}(K)$ and let $\phi \in \text{Hom}(K, \text{Aut}(H))$
 $[(\psi_1 \circ \psi_2, \mu_1 \circ \mu_2)\phi](k) = \psi_1 \circ \psi_2 \circ (\phi(\mu_2^{-1} \circ \mu_1^{-1}(k))) \circ \psi_2^{-1} \circ \psi_1^{-1}$
 $((\psi_1, \mu_1)[(\psi_2, \mu_2)\phi])(k) = (\psi_1, \mu_1)(\psi_2 \circ (\phi(\mu_2^{-1}(k))) \circ \psi_2^{-1}) =$
 $\psi_1 \circ \psi_2 \circ (\phi(\mu_2^{-1} \circ \mu_1^{-1}(k))) \circ \psi_2^{-1} \circ \psi_1^{-1}$
Thus this is an action
- Assume ϕ, ϕ' are in the same orbit for this action this means
there exists a $(\psi, \mu) \in \text{Aut}(H) \times \text{Aut}(K)$ s.t. $\phi'_k = (\psi, \mu)\phi_k = \psi \circ \phi_{\mu^{-1}(k)} \circ \psi^{-1}$
Let $\varphi : H \rtimes_{\phi} K \rightarrow H \rtimes_{\phi'} K$ s.t. $\varphi(h_1, k_1) = (\psi(h), \mu(k))$

Homomorphism: Let $(h_1, k_1), (h_2, k_2) \in H \rtimes_{\phi} K$

$$\phi[(h_1, k_1)(h_2, k_2)] = \phi(h_1 \phi_{k_1}(h_2), k_1 k_2) = (\psi(h_1 \phi_{k_1}(h_2)), \mu(k_1, k_2))$$

$$\begin{aligned} \phi(h_1, k_1) \phi(h_2, k_2) &= (\psi(h_1), \mu(k_1))(\psi(h_2), \mu(k_2)) = (\psi(h_1) \phi'_{\mu(k)}(\psi(h_2)), \mu(k_1) \mu(k_2)) \\ &= (\psi(h_1) \psi \circ \phi_{\mu^{-1}(\mu(k_1))} \circ \psi^{-1}(\psi(h_2)), \mu(k_1 k_2)) = (\psi(h_1 \phi_{k_1}(h_2)), \mu(k_1 k_2)) \end{aligned}$$

Injective: Let $(h_1, k_1), (h_2, k_2) \in H \rtimes_{\phi} K$ s.t. $\phi(h_1, k_1) = \phi(h_2, k_2)$

$$\phi(h_1, k_1) = \phi(h_2, k_2)$$

$$(\psi(h_1), \mu(k_1)) = (\psi(h_2), \mu(k_2))$$

thus $\psi(h_1) = \psi(h_2)$ and $\mu(k_1) = \mu(k_2)$ since these are automorphisms $h_1 = h_2$ and $k_1 = k_2$

thus $(h_1, k_1) = (h_2, k_2)$

Surjective: Let $(h, k) \in H \rtimes_{\phi'} K$ consider $(\psi^{-1}(h), \mu^{-1}(k))$

$$\phi(\psi^{-1}(h), \mu^{-1}(k)) = (h, k)$$

Thus $H \rtimes_{\phi} K \cong H \rtimes_{\phi'} K$

Problem 5

Used some latex from class notes.

Let n_3 be the number of 3-Sylows then by Sylow Theorem 3, $n_3 \mid 7$ and $n_3 \equiv 1 \pmod{3}$. Thus, $n_3 = 1$ or $n_3 = 7$

Let n_7 be the number of 7-Sylows then by Sylow 3, $n_7 \mid 3$ and $n_7 \equiv 1 \pmod{7}$, thus $n_7 = 1$.

If $n_3 = 1$ then this implies H is a unique 3-Sylow subgroup of G , and thus it is normal by Sylow 2 this applies for K the unique 7-Sylow subgroup of G .

Consider HK : Since H, K are both normal HK is a subgroup of G . Also, $H \cup K$ has 9 elements of the 21 elements in G , thus by Lagrange's theorem the only other order of elements outside $H \cup K$ would have order 21, thus $G \cong C_{21}$

If $n_3 = 7$. We still have that the unique 7-Sylow subgroup $K = \{e, x, x^2, \dots, x^6\}$ which is normal in G , and we can fix an arbitrary 3-Sylow subgroup $H = \{e, y, y^2\}$.

Since K is normal this implies $xyx^{-1} = x^i$ thus

$$\begin{aligned} y^2xy^{-2} &= y(yxy^{-1})y^{-1} = yx^iy^{-1} = (yxy^{-1})^i = (x^i)^i = x^{(i^2)} \\ x &= y^3xy^{-3} = y(y^2xy^{-2})y^{-1} = y(x^{(i^2)})y^{-1} = (x^{(i^2)})^i = x^{(i^3)} \end{aligned}$$

From this we can see that $i^3 \equiv 1 \pmod{7}$, reducing the possibilities to $i \in \{1, 2, 4\}$.

- If $i = 1$ then $yx = xy$ thus consider xy

$(xy)^2 = x^2y^2$	$(xy)^9 = x^2$	$(xy)^{16} = x^2y$
$(xy)^3 = x^3y^3 = x^3$	$(xy)^{10} = x^3y$	$(xy)^{17} = x^3y^2$
$(xy)^4 = x^4y$	$(xy)^{11} = x^4y^2$	$(xy)^{18} = x^4$
$(xy)^5 = x^5y^2$	$(xy)^{12} = x^5$	$(xy)^{19} = x^5y$
$(xy)^6 = x^6$	$(xy)^{13} = x^6y$	$(xy)^{20} = x^6y^2$
$(xy)^7 = y$	$(xy)^{14} = y^2$	$(xy)^{21} = e$
$(xy)^8 = xy^2$	$(xy)^{15} = x$	

Thus $\text{ord}(xy) = 21$ thus $G \cong C_{21}$

- If $i = 2$

Thus $yxy^{-1} = x^2$

Thus $G = \langle x, y \mid x^7 = y^3 = e, yxy^{-1} = x^2 \rangle$

- If $i = 4$

Thus $yxy^{-1} = x^4$

thus $y^2xy^{-2} = x^{16} = x^2$

Thus let $\varphi : G \rightarrow \langle a, b \mid a^7 = b^3 = e, bab^{-1} = a^2 \rangle = A$ s.t. $\varphi(x) = a$ and $\varphi(y^2) = b$

Homomorphism: We only need to check for we sent same order to same order which we did

$$\text{and that } \varphi(y^2)\varphi(x) = \varphi(x)^2\varphi(y^2)$$

$$\varphi(y^2)\varphi(x) = ba = ba^2 = \varphi(x)^2\varphi(y^2)$$

Injective: Let $g \in \ker(\varphi)$

Since we did not send any generators of G to the identity element this means that our kernel is trivial thus φ is injective

Surjective: Let $g \in A$ thus $g = a^i b^j$ for $0 \leq i < 7$ and $0 \leq j < 3$

Consider $x^i y^{2j}$

$$\varphi(x^i y^{2j}) = \varphi(x)^i \varphi(y^2)^j = a^i b^j = g$$

Thus $G \cong A$

Problem 6

Only have to check generators for conjugacy

Consider: x^n

$yx^ny^{-1} = x^{2n}$ thus

Class 1: $\{x, x^2, x^4\}$

Class 2: $\{x^3, x^5, x^6\}$

Consider: y

$x^i y x^{-i} = y x^{2i} x^{-i} = y x^i$ for $i \in \{0, \dots, 6\}$

Class 3: $\{y, yx, yx^2, yx^3, yx^4, yx^5, yx^6\}$

Consider: y^2

$x^i y^2 x^{-i} = y x^{2i} y x^{-i} = y^2 x^{3i}$ for $i \in \{0, \dots, 6\}$

Class 4: $\{y^2, y^2x, y^2x^2, y^2x^3, y^2x^4, y^2x^5, y^2x^6\}$

Class 5: $\{e\}$

Thus $|G| = 1 + 3 + 3 + 7 + 7 = 21$

Problem 7

Let G be order $30 = 2 \times 3 \times 5$.

By Sylow 1 there exists 2-Sylow subgroup, 3-Sylow subgroup, and 5-Sylow subgroup.

Let n_5 be the number of 5-Sylow subgroups the by Sylow 3, $n_5 \mid 6$ and $n_5 \equiv 1 \pmod{5}$

Thus $n_5 = 1, 6$

Let n_3 be the number of 3-Sylow subgroups the by Sylow 3, $n_3 \mid 10$ and $n_3 \equiv 1 \pmod{3}$

Thus $n_3 = 1, 10$

If $n_5 = 6$ and $n_3 = 10$ then this means there are $2 \cdot 10 = 20$ elements of order 3

this leaves 9 elements to be order 5 however there are $4 \cdot 6 = 24$ elements of order 5 thus these cannot be true at the same time

Thus n_5 or n_3 is equal to 1.

If $n_5 = 1$ let H be the unique 5-Sylow. Thus, H is normal in G , and let K be a 3-Sylow.

Since H is normal this implies HK is a subgroup of G . Also since H has only elements of order 5, 1 and K only has elements of order 3, 1 this implies $K \cap H = \{e\}$ thus $|HK| = \frac{|H||K|}{|H \cap K|} = 15$ thus $[G : HK] = 2$ by lagrange's theroem. Thus, since HK is index 2 this means it is normal in G

If $n_3 = 1$ do the same thing as above, and you will get a normal subgroup of order 15.

Let L be that normal subgroup of order 15 since it is order 15 proven in class this can only be isomorphic C_{15} thus let $L = \langle x \rangle$ for $x \in L$

Let U be a 2-Sylow subgroup since it is order of a prime we know it is cyclic thus let $U = \langle y \rangle$ for $y \in U$

Since L is normal this implies $xyx^{-1} = x^i$ thus

$x = y^2xy^{-2} = y(yxy^{-1})y^{-1} = yx^iy^{-1} = x^{2i}$ thus

$i^2 \equiv 1 \pmod{15}$

$i \in \{1, 4, 11, 14\}$

- If $i = 1$ then $yx = xy$ thus consider xy

$$(xy)^2 = x^2y^2 = x^2$$

$$(xy)^3 = x^3y^3 = x^3y$$

$$(xy)^4 = x^4$$

$$(xy)^5 = x^5y$$

$$(xy)^6 = x^6$$

$$(xy)^7 = x^7y$$

$$(xy)^8 = x^8$$

$$(xy)^9 = x^9y$$

$$(xy)^{10} = x^{10}$$

$$(xy)^{11} = x^{11}y$$

$$(xy)^{12} = x^{12}$$

$$(xy)^{13} = x^{13}y$$

$$(xy)^{14} = x^{14}$$

$$(xy)^{15} = y$$

$$(xy)^{16} = x$$

$$(xy)^{17} = x^2y$$

$$(xy)^{18} = x^3$$

$$(xy)^{19} = x^4y$$

$$(xy)^{20} = x^5$$

$$(xy)^{21} = x^6y$$

$$(xy)^{22} = x^7$$

$$(xy)^{23} = x^8y$$

$$(xy)^{24} = x^9$$

$$(xy)^{25} = x^{10}y$$

$$(xy)^{26} = x^{11}$$

$$(xy)^{27} = x^{12}y$$

$$(xy)^{28} = x^{13}$$

$$(xy)^{29} = x^{14}y$$

$$(xy)^{30} = e$$

Thus $G \cong C_{30}$

- If $i = 14$ then $xyx^{-1} = x^{14} \implies yx = x^{14}y$
thus $G = \langle x, y \mid x^{15} = y^2 = e, yx = x^{14}y \rangle \cong D_{15}$
- If $i = 4$ then $xyx^{-1} = x^4$ $G = \langle x, y \mid x^{15} = y^2 = e, yxy^{-1} = x^4 \rangle$
- If $i = 11$ then $xyx^{-1} = x^{11}$ $G = \langle x, y \mid x^{15} = y^2 = e, yxy^{-1} = x^{11} \rangle$